Left-Branching Transformers Excel at Right-Branching Languages: Data Shapes Word Order Preferences in Language Models

Varvara Arzt[1], **Allan Hanbury**[1], **Terra Blevins**[2]

[1]Faculty of Informatics, TU Wien
[2]Khoury College of Computer Sciences, Northeastern University

Correspondence: varvara.arzt@tuwien.ac.at

Abstract

We systematically compare word order preferences in decoder-only language models across 192 artificial languages and typologically diverse natural languages. On artificial languages, models exhibit a left-branching preference that aligns with neither natural language universals nor human word order learning biases. On natural languages, monolingual models show no clear base word order bias at small scales, but as data grows, a preference for right-branching subject-verb-object (SVO) languages emerges while SOV falls behind despite being the most frequent order cross-linguistically. This SVO advantage extends to multilingual models and correlates with language resource level and data quality rather than word order. Thus, the same architecture exhibits opposite preferences on artificial and natural languages, establishing that word order biases observed in practice are data-driven. Since highly-resourced languages are overwhelmingly SVO, these biases risk gradually reducing word order diversity, particularly in languages that productively use multiple word orders, with the widespread adoption of LLMs.[1]

1 Introduction

Word order is one of the most fundamental dimensions of cross-linguistic variation. Languages vary widely in how they linearise both clause-level constituents (e.g., SVO in English, SOV in Japanese) and internal constituent orderings, such as the placement of adpositions and adjectives relative to their heads; however, many of these ordering choices tend to correlate (Greenberg, 1963; Dryer, 1992; Dryer and Haspelmath, 2013; Verkerk et al., 2026) with, for instance, SOV languages typically using postpositions and SVO languages prepositions, placing languages on a scale of *directionality* (Figure 1a) from *left-branching*, where dependents precede their heads (e.g., postpositions), to *right-branching*, where heads come first (e.g., prepositions). Beyond dominant orders, many languages exhibit productive word order variation that serves discourse-pragmatic functions such as topicalisation and focus marking (Hüll and Dobrovoljc, 2025). As language models (LMs) are increasingly

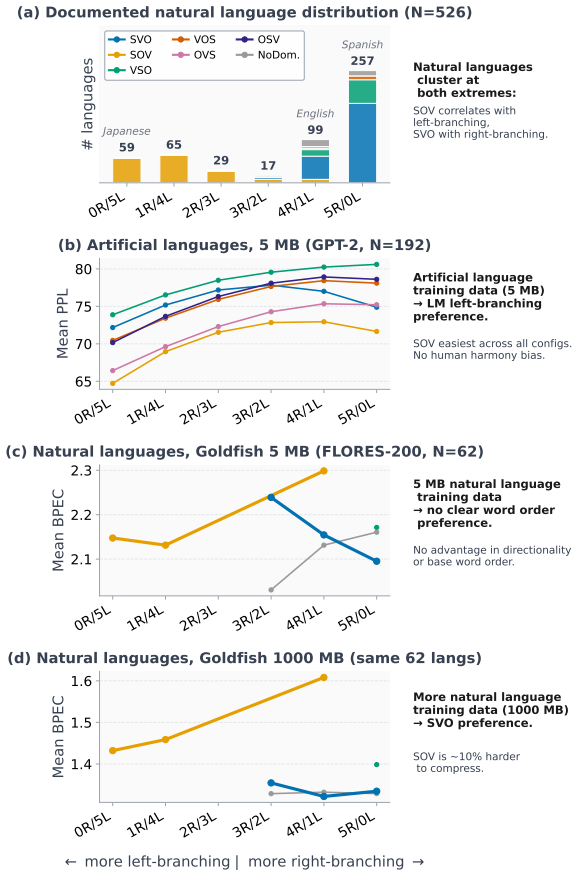


Figure 1: Head directionality vs. model performance across attested, artificial, and natural languages. x-axis: right-branching switch count (§3.1), from fully left-branching (0R/5L) to fully right-branching (5R/0L). ↓ = better in (b)–(d); colours denote base word order (legend in (a)). **(a)** Attested language frequency by base word order. **(b)** GPT-2 on artificial languages (PPL). **(c, d)** Monolingual models on natural languages at two training data scales (BPEC; §3.4).

[1]Code and data are available at https://github.com/kleines-gespenst/word-order-preferences

Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing, pages 56198–56219
October 24-29, 2026 ©2026 Association for Computational Linguistics

deployed across typologically diverse languages, a natural question arises: do these models develop systematic word order preferences, and if so, are such preferences architectural or data-driven? This question is especially pressing due to the potential for LLM-mediated language homogenisation (Kew et al., 2024; Fitterer et al., 2025) and LLM bias towards more formal writing styles (Abdulhai et al., 2026), which tend to favour rigid word orders.

Prior work has used artificial languages generated from formal grammars (e.g., White and Cotterell, 2021) and synthetic variants of natural languages (e.g., Ravfogel et al., 2019) to study the word order preferences of LMs. However, artificial languages lack the semantic, morphological, and discourse-level complexity of natural languages, so findings from controlled settings may not generalise; conversely, evaluation on natural languages alone cannot isolate word order from confounds such as morphology, data quality, and script. We combine both approaches at multiple training data scales to address this gap, providing three methodological contributions:

(1) We train models on 192 artificial languages spanning all six base word orders as well as 32 internal constituent orderings, analysing directionality preferences, alignment with cross-linguistic universals, and constituent-level surprisal at multiple training data scales.

(2) We similarly evaluate monolingual and multilingual models trained on typologically diverse natural languages, analysing how word order preferences and constituent surprisal scale with growing data.

(3) By mapping artificial language configurations to natural language typological features and comparing models trained at matched data scales across both paradigms, we unify artificial and natural evaluations of word order preference into a single, controlled comparison, allowing us to disentangle architectural and data-driven biases.

We find that on artificial languages, models exhibit a left-branching preference (Figure 1b) that aligns with neither cross-linguistic universals nor human learners' preference for consistent directionality (Culbertson and Newport, 2015). Constituent surprisal analyses suggest this preference may partly reflect the entropy structure of a semantics-free grammar rather than a model-inherent architectural prior. However, these base word order rankings can shift with data volume, and directionality preferences reverse early in training.

In contrast, while no word order preference is visible at 5 MB (Figure 1c) for natural languages, we find that SVO languages are easiest and SOV hardest in monolingual models at 1000 MB (Figure 1d), implicating data quality rather than quantity alone. This SVO advantage extends to multilingual models and disappears only when this word order is dominated by very-low-resource languages, as in BLOOM (BigScience Workshop et al., 2023).

These findings show that, in practice, models consistently favour the word orders of highly-resourced languages, particularly at scale. Since highly-resourced languages are overwhelmingly SVO (e.g., English and Chinese; Appendix A.4), this creates a precondition for *silent typological homogenisation*: a gradual reduction of word order diversity driven by imbalanced training data. Our results suggest that this precondition is real, robust across model types and training regimes, and in principle addressable through data curation.

2 Related Work

LM word order preferences have mainly been studied through artificial languages and synthetic variants of natural languages.

White and Cotterell (2021) train LMs on languages generated from a probabilistic context-free grammar (PCFG) varying in branching directionality, finding a preference for left-branching configurations and best performance on both widely attested SOV and rarely attested OVS, suggesting no alignment with typological frequency; our artificial language experiments build upon their approach (§3.1). Kuribayashi et al. (2024) extend this to cognitively-motivated LMs, finding that models with parsing strategies and memory limitations achieve lower perplexity (PPL) on typologically frequent word orders. El-Naggar et al. (2025b) adopt Generalized Categorial Grammar, covering all six S/V/O linearisations, and find that typologically plausible orders facilitate length generalisation. In emergent communication, Lian et al. (2023) show that neural agents converge from an equal-probability mixed-order language toward a single dominant order through communicative pressure alone, an effect that strengthens with larger agent group size (Lian et al., 2024).

A complementary line of work creates synthetic variants of attested languages. Ravfogel et al. (2019) create synthetic English variants with altered word order and case marking, finding that RNN agreement prediction reflects recency bias. Bisazza et al. (2021) similarly alter word order flexibility and case marking in English to study neural machine translation, finding that translating flexible-order languages is harder in low-resource settings. Kallini et al. (2024) show that GPT-2 assigns higher PPL to English variants with disrupted hierarchical structure, though Ziv et al. (2026) find this does not generalise across nine diverse languages. Clark et al. (2023) find that real word orders distribute information more uniformly than counterfactual alternatives, with the effect strongest for SVO. In concurrent work, Xu et al. (2026) find that disharmonic (i.e., typologically inconsistent in directionality) variants of English and Japanese based on five Greenbergian universals[2] are learned more slowly but reach comparable final performance; we test a partly overlapping set of universals (§3.4).

No prior work, to our knowledge, combines controlled artificial and natural language experiments to analyse word order preferences; we do so here.

3 Methodology

To disentangle potential architectural biases from effects of training data, we perform parallel experiments on artificial and natural languages, using statistics on attested[3] languages as a reference for whether model preferences correlate with human biases. Artificial languages allow us to analyse all possible word order permutations across syntactic constructions in isolation, avoiding confounds from typological dimensions like morphological agreement (White and Cotterell, 2021). Experiments with natural languages address whether findings on artificial languages generalise, and help establish whether word order preferences can be reliably studied with artificial languages (El-Naggar et al., 2025a; Ziv et al., 2026).

3.1 Artificial Languages

We generate artificial languages building upon White and Cotterell (2021). Each language is defined by a base S/V/O order and five binary head-direction switches controlling the ordering of complement clauses, complementisers, adpositions, adjectives, and relative clauses (Figure 2). Each switch selects between an **L** (left-branching, head-final) variant, in which the dependent precedes the head, and an **R** (right-branching, head-initial) variant, in which the head precedes the dependent. A base PCFG with a Zipf-distributed lexicon generates bracketed parse trees; main clause constituents are then deterministically reordered and head-direction switches applied, so that all 192 variants share identical derivation probabilities and differ only in word order. Figure 3 illustrates this with the same sentence under `SOV-LLLLL` (≈ Japanese) and `SVO-RRRLR` (≈ English): the hierarchical structure is preserved while the surface linearisation changes.

Each language is encoded as `[Base]-[5-char]`, where the base specifies the S/V/O order and each character selects a binary parameter: `L` = left-branching, `R` = right-branching.

Switch	L (left-branching)	R (right-branching)
1 VP_Comp	$S_{Comp}\ Verb_{Comp}$	$Verb_{Comp}\ S_{Comp}$
2 Comp	$S\ Comp$	$Comp\ S$
3 PP	$NP\ Postp$	$Prep\ NP$
4 NP	$Adj\ Noun$	$Noun\ Adj$
5 Rel	$VP\ Rel\ Noun$	$Noun\ Rel\ VP$

Example: `SOV-LLLLL` ≈ **Japanese**

Feature	Value
Base (SOV)	Subject–Object–Verb
Switch 1 = L	complement clause before verb
Switch 2 = L	clause before complementiser
Switch 3 = L	postpositions
Switch 4 = L	adjective before noun
Switch 5 = L	prenominal relative clause

6 base orders × 2^5 configurations = **192 languages**.

Figure 2: Language encoding scheme

We extend White and Cotterell (2021) in three ways: (1) we add an explicit switch for clausal complement order, which White and Cotterell (2021) tie to nominal object order, making them inseparable by design;[4] (2) we add a six-way base word order parameter covering all six S/V/O linearisations, as does El-Naggar et al. (2025b) but via a different formalism. Combining six base orders with 2^5 switch configurations yields **192 artificial languages** with

[2]Cross-linguistic generalisations about word order co-occurrences (Greenberg, 1963), e.g., SOV ↔ postpositions.

[3]Documented natural languages

[4]Nominal and clausal object placement are typologically correlated (Dryer, 1992) but not always identical; e.g., the Agob-Ende-Kawam language has preverbal nominal but postverbal clausal objects (Skirgård et al., 2023, GB135).

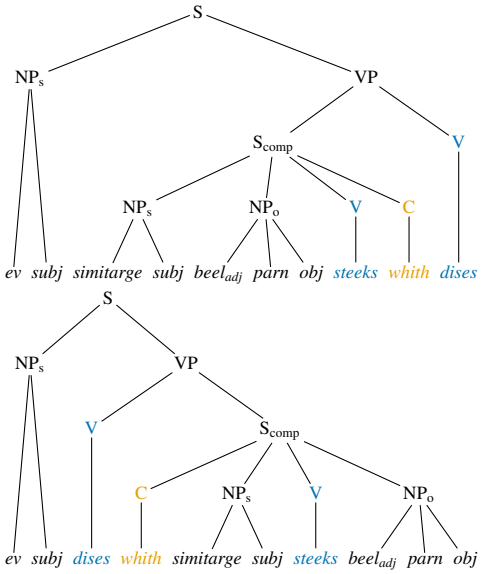


Figure 3: Same sentence as SOV-LLLLL (≈ Japanese, top) and SVO-RRRLR (≈ English, bottom): same hierarchical structure, but word order differs. Verbs & complementiser highlighted; *subj*/*obj* = overt case markers.

fixed word order; and (3) we scale the vocabulary from ∼1,400 to 50k pseudowords generated by Wuggy (Keuleers and Brysbaert, 2010) from the most frequent English words in wordfreq (Speer, 2022), with Zipfian sampling weights (α=1.0) per part-of-speech class (details in Appendix A.1).

3.2 Natural Languages

Word Order Clustering To enable comparison between artificial and natural language experiments, we classify each natural language using the same encoding scheme applied to our artificial languages. We obtain word order labels from the typological databases WALS (Dryer and Haspelmath, 2013), Grambank (Skirgård et al., 2023), and APiCS (Michaelis et al., 2013) and map them to our artificial language labels (e.g., English → SVO-RRRLR, Japanese → SOV-LLLLL). We supplement database labels with manual verification grounded in peer-reviewed publications devoted to individual languages, and use `lang2vec` (Littell et al., 2017) and URIEL+ (Khan et al., 2025) for language clustering and feature extraction. Languages with non-binary or missing values for a given feature were excluded from our analysis. Full feature mapping procedure and coverage statistics are given in Appendix A.2.

We note that typological database classifications are typically based on grammar descriptions rather than corpus frequencies (Levshina et al., 2023; Hüll and Dobrovoljc, 2025), and a single categorical label may obscure gradient word order variation (Levshina et al., 2023); see § Limitations for further details.

Data We evaluate on two datasets. Our primary evaluation set is FLORES-200 (NLLB Team et al., 2022), which provides parallel sentences across a typologically broad set of languages. We also evaluate on the Parallel Universal Dependencies (PUD) treebanks (Zeman et al., 2017), whose languages are all covered by FLORES-200, which both verify our FLORES-200 findings and provide gold-standard syntactic annotations for the per-token surprisal analysis, complementing automatic Stanza (Qi et al., 2020) parses on FLORES-200. Full list of languages used in §5 is given in Appendix A.2.

3.3 Models

For artificial languages, we train a separate GPT-2-style decoder-only model on each of the 192 languages to evaluate whether word order preferences arise from the transformer architecture itself. For natural languages, we primarily focus on monolingual Goldfish models (Chang et al., 2026), GPT-2-based models with a 50k-token vocabulary, scaled in parameter count with training data volume, trained on one of up to 350 languages in a maximally comparable setup at four training data volumes (5 MB, 10 MB, 100 MB, and 1000 MB of content-equivalent text per language after byte premium scaling; Chang et al., 2026)[5], enabling systematic analysis of how word order preferences develop with growing amounts of data. To verify whether these patterns generalise to multilingual training, we additionally evaluate mGPT (1.3B and 13B parameters, ∼400B tokens across 61 languages with no exact per-language statistics available; (Shliazhko et al., 2024)), BLOOM (560M, 3B, and 7.1B parameters, 341B tokens across 46 languages, median 527 MB per language; (BigScience Workshop et al., 2023; Laurençon et al., 2022)) and XGLM (564M and 1.7B parameters, ∼500B tokens across 30 languages with exponential upsampling of low-resource languages; (Lin et al., 2022)). All three multilingual models were designed to broaden language coverage beyond English-centric corpora and document their training data composition, allowing us to verify which languages each

[5]Byte premium scaling adjusts raw data sizes by UTF-8 byte ratio relative to English (Arnett et al., 2024): e.g., the 5 MB Burmese model was trained on 25 MB raw text (`https://huggingface.co/goldfish-models/mya_mymr_5mb`).

model was trained on and to assess potential contamination with FLORES-200 and PUD.

3.4 Evaluation

We evaluate word order preferences at two levels: clause-level S/V/O linearisation and internal constituent order (the five switches in Figure 2), aggregating the latter into a directionality scale.

For artificial languages, we report test **perplexity** (PPL). Following Kallini et al. (2024) and Ziv et al. (2026), we additionally analyse training dynamics by tracking test perplexity every 50 steps and computing the area under the PPL curve (AUC); this is not possible for Goldfish models as no intermediate checkpoints are available. We also test whether model preferences align with a subset of **Greenbergian universals** (Greenberg, 1963) testable in our setup: subject-object ordering, base word order and adposition type, adjective placement, and verb-object order with relative clause placement (Dryer, 1992) (details in Appendix A.3).

For natural languages, where different tokenisers make raw PPL incomparable across languages, we use **bits-per-English-character** (BPEC; Appendix A.3) as defined in Cotterell et al. (2018). Throughout, a language is *easier* (*harder*) for a model if it achieves lower (higher) PPL/BPEC. For both artificial and natural languages, we evaluate per-token **surprisal** (Wilcox et al., 2023) to analyse whether tokens with different syntactic functions are differentially predictable across word orders.

4 Experimental Setup

4.1 Artificial Languages

We train GPT-2-style decoder-only models using the HuggingFace Transformers library[6] (`GPT2LMHeadModel`) at three training data volumes: the original White and Cotterell (2021) setup (∼0.5 MB, 10K sentences), 5 MB, and 10 MB. As 0.5 MB results are comparable to 5 MB, and we aim for comparability with the monolingual Goldfish models, we focus on the 5 MB and 10 MB settings.

The model architecture follows the Goldfish configuration at comparable sizes: 4 layers and 8 attention heads (Chang et al., 2026). Unlike White and Cotterell (2021), who used a whitespace tokeniser (under which the vocabulary is effectively arbitrary), we train a BPE word-boundary SentencePiece tokeniser (Kudo and Richardson, 2018) with a vocabulary size of 15k (details in Appendix A.1). Following White and Cotterell (2021), we train each model on 10 fully disjoint train/dev/test splits generated from independent data and report averaged test perplexity to ensure robustness of results. Implementation details, hyperparameters and compute cost reported in Appendix A.1.

4.2 Natural Languages

We evaluate Goldfish models, available on HuggingFace,[7] at all four training data sizes and the multilingual models introduced in §3.3 on FLORES-200 (NLLB Team et al., 2022) (Appendix A.2) and PUD (Zeman et al., 2017). Syntactic annotations for the surprisal analysis come from Stanza (Qi et al., 2020) on FLORES-200 and gold UD annotations on PUD. For multilingual models, we restrict evaluation to the subset of each model's training languages overlapping with FLORES-200 (language counts per model in Table 2).

5 Results

We compare LM word order preferences on artificial languages (§5.1), where models show a strong left-branching preference misaligned with cross-linguistic universals, and natural languages (§5.2), where training data composition increasingly shapes model preferences.

5.1 Artificial Languages

On artificial languages, models consistently prefer left-branching configurations, conflicting with cross-linguistic universals, and base word order rankings shift unstably with data volume.

Directionality Preferences Figure 1b shows that the mean PPL across settings increases consistently with the number of right-branching switches in an artificial language across all six base word orders, with only slight decreases in PPL for majority right-branching settings in SVO and SOV languages. This preference for left-branching configurations corroborates prior work (White and Cotterell, 2021; Kuribayashi et al., 2024; El-Naggar et al., 2025a).

However, the top-ranked configuration for each base order has no close natural language counterpart,[8] suggesting that transformers do not mirror human word order learning biases. In contrast, attested languages cluster at both extremes of the

[6] https://huggingface.co/docs/transformers

[7] https://huggingface.co/goldfish-models

[8] A partial exception is SVO-RRRLL (≈ Mandarin Chinese), though the prenominal relative clause (switch 5 = L) is very rare in VO languages (Dryer, 1992; Comrie, 2008).

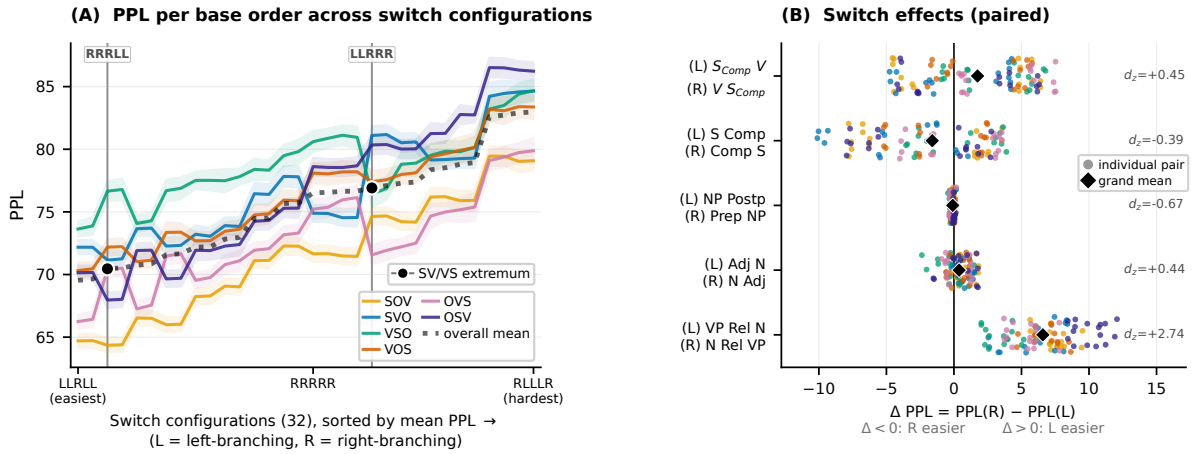


Figure 4: LM preferences on 192 artificial languages (6 base orders × 32 branching configurations), averaged over 10 runs on 5 MB data. **(A)** PPL per base order across configurations, sorted by across-order mean (dotted line). Shaded bands: ±1 SD. Vertical markers (RRRLL, LLRRR): SV and VS base orders respond inversely to clause-level switches. **(B)** Paired switch effects (ΔPPL) for configuration pairs differing only in the target switch. ♦: overall mean, where > 0 = left-branching is preferred. Switch effect sizes d_z are all significant ($p < .001$).

directionality scale (Figure 1a), consistent with the human harmony bias toward consistent directionality (Culbertson and Newport, 2015).

The magnitude of the left-branching preference also varies substantially across individual switches (Figure 4B). Switch 5 (relative clause) dominates: all sixteen left-branching relative clause configurations are easier than the right-branching ones, at ΔPPL of 4–7 points on 5 MB. This strong preference for prenominal relatives runs counter to the typological distribution, where postnominal relatives outnumber prenominal ones approximately 4:1 (Dryer, 1992; Dryer and Haspelmath, 2013). Other switches show smaller but consistent trends; for example, Switch 4 (pre- or postnominal adjectives) shows a moderate preference for adjectives preceding the noun. All five paired t-tests on the effect of internal constituent switches are significant even after a strict Holm–Bonferroni correction to remove false positives ($p < .001$). More broadly, switches that reorder clause-level constituents (relative clauses, complement clauses) produce larger perplexity differences than switches that rearrange phrase-internal elements (adjective–noun, adposition–NP), suggesting greater sensitivity to the direction of long-range dependencies.

Base Word Order Model preferences of base word orders are not stable across training data volumes and also do not reflect typological frequency (Appendix Figure 11). At 5 MB, SOV ranks first and VSO last in mean PPL across configurations (Figure 4A), consistent with White and Cotterell (2021). At 10 MB, the three VS orders rank 1–3 and the three SV orders rank 4–6, with SVO the hardest (Appendix Figure 9), which is opposite the trend we see on natural languages (where SVO is easiest to model at scale; §5.2).

Internal constituents also interact with base word order, with individual base order curves crossing over for specific switch settings (Figure 4A): SV bases (SVO, SOV, OSV) and VS bases (VSO, VOS, OVS) respond differently to individual switches. Right-branching verb complements and complementisers (switches 1–2 = R) selectively benefit SV bases by up to 5.5 PPL points (vertical marker RRRLL in Figure 4A), while right-branching relative clauses (switch 5 = R) reverse this, favouring VS bases by up to 3.8 points (vertical marker LLRRR). This observed SV/VS split does not align with the VO/OV distinction underlying most implicational word order universals (Greenberg, 1963; Dryer, 1992; Dryer and Haspelmath, 2013), and direct tests of typologically expected combinations confirm this: the tested universals are either not reflected (adposition, adjective placement) or significant in the opposite direction (subject–object order and relative clause placement, at 10 MB). These results indicate that the inductive biases demonstrated by transformers on artificial languages do not match cross-linguistic word order universals.

Training Dynamics The model's directionality preference shifts in the earliest phase of training,

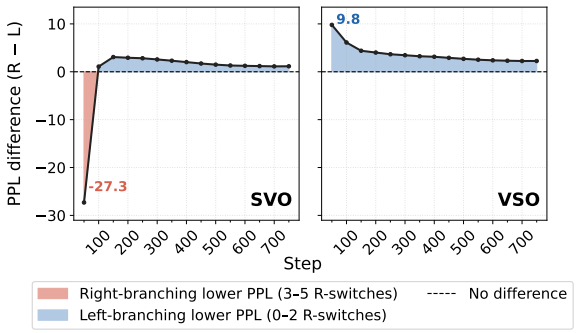


Figure 5: PPL Δ (right − left branching) during training on 5 MB artificial data. **SVO**: right-branching is initially easier before reversing to a left-branching advantage by step 100. **VSO**: left-branching always dominates.

Base order	5 MB	10 MB	100 MB	1 GB
SVO (32)	**2.13** (.15)	**2.00** (.15)	**1.55** (.08)	**1.32** (.06)
SOV (22)	**2.13** (.10)	2.02 (.10)	1.64 (.11)	1.45 (.12)
VSO (3)	2.20 (.07)	2.06 (.09)	1.64 (.06)	1.42 (.06)
NoDom (11)	**2.13** (.11)	**2.00** (.10)	1.56 (.08)	1.33 (.05)
SVO–SOV δ	0.00	−0.10	-0.60^{***}	-0.76^{***}

At 5 MB: SVO = 2.1288, SOV = 2.1289, NoDom = 2.1284.

Table 1: Median BPEC and (interquartile range; IQR) for base word orders of 68 monolingual Goldfish models (evaluated on FLORES-200, lang. count in parentheses). ↓ = better, **bold** = best per column. Bottom row: SVO–SOV effect size (Cliff's δ); order preference is absent at 5 MB (Mann–Whitney U=353, p=.99) and large by 1 GB (U=84, $p<10^{-5}$). $^{***}p<.001$.

again following the SV/VS split (Figure 5 for SVO and VSO, with other orders following this trend in Appendix Figure 10). For SV base orders, right-branching configurations are initially easier (by up to 28 PPL points) before fully reversing to a left-branching advantage by step 100 (epoch ∼2.6); for VS orders, left-branching dominates from the first checkpoint. Throughout training, the internal constituent order switches rather than the base S/V/O order drive the PPL spread, with convergence speed and final PPL strongly correlated ($\rho > 0.89$; Appendix Figure 8). Moreover, the early reversal for SV languages echoes the preference shift observed by El-Naggar et al. (2025a) and indicates that directionality preferences may be sensitive to training dynamics and data quantity, which should be considered when comparing results across studies.

5.2 Natural Languages

While LMs show no word order preference in small training data scales on natural languages, an SVO advantage emerges at scale in both mono- and multilingual models, correlating with language resource level and data quality rather than word order type.

Directionality Preferences Figure 1c demonstrates that at 5 MB, LMs trained on natural languages exhibit no clear directionality preference; however, by 1000 MB, right-branching SVO languages become easiest (Figure 1d; intermediate scales in Appendix Figure 6).[9] This may reflect the same left-branching inductive bias observed in artificial languages, overridden by data effects in natural languages as training data grows.

Base Word Order Preferences across Scale This shift across data scales is also visible at the level of base word order. At 5 MB, monolingual Goldfish models show no preference: SVO, SOV, and NoDominant all centre around BPEC 2.13, with no significant difference between orders (Table 1). As data grows, an SVO advantage emerges and strengthens monotonically (SVO–SOV Cliff's δ=−0.76 at 1 GB): by 1 GB, SOV incurs ∼10% higher compression cost (ΔBPEC ≈ 0.13), despite being the most frequent order cross-linguistically (Appendix Figure 11). All four model sizes are evaluated on the same 68 languages, so the shift is driven solely by training data volume, not language selection. The same pattern holds on PUD (details in Appendix Table 7).

Because word order correlates with other language properties, the observed SVO advantage may also be an artefact of a confound such as morphological complexity (Arnett and Bergen, 2025) rather than a genuine word order preference. To test this, we fit a linear mixed-effects model on the BPEC of the SVO/SOV languages in Figure 1c,d, with a word order × size interaction, per-language random intercepts, and three covariates: morphological complexity (subword MATTR; Tatariya et al. 2025), language resourcedness (Joshi et al., 2020), and training data composition (OSCAR web-crawl vs. curated share; Appendix A.5). With all three covariates included, the growth of the SVO–SOV gap with data survives every control (β: +0.049 → +0.038, $p\leq.001$; Appendix Table 6). Language resourcedness is the only covariate with an observable effect on this growth

[9] Here we restrict to the 68 languages with Goldfish models at all four training data sizes. When all available languages at each size are included (Appendix Figure 7), SVO even underperforms SOV at small scales.

Model	SVO	SOV	NoDom	δ
BLOOM-560m	3.62 (2.86)	**2.59** (.59)	—	$+.26$
BLOOM-3b	2.99 (2.20)	**1.88** (.34)	—	$+.27$
BLOOM-7b1	2.81 (2.06)	**1.82** (.29)	—	$+.25$
XGLM-564M	**1.46** (1.06)	1.89 (4.69)	1.47 (.63)	$-.49^*$
XGLM-1.7B	**1.41** (.88)	1.78 (5.12)	1.42 (.54)	$-.53^*$
mGPT-1.3B	**1.53** (.28)	1.74 (.25)	1.64 (.31)	$-.43^*$
mGPT-13B	**1.44** (.25)	1.68 (.18)	1.54 (.24)	$-.44^*$

Table 2: Median FLORES-200 BPEC and (IQR) by base word order for multilingual models. Language counts (SVO/SOV/NoDom): BLOOM 28/13/0, XGLM 14/10/3, mGPT 25/22/9. ↓ = better, **bold** = best per model. Last column: SVO–SOV effect size (Cliff's δ); SVO is preferred by XGLM and mGPT ($\delta<0$) but not BLOOM ($\delta>0$), whose SVO group is very-low-resource and heterogeneous (wide SVO IQR). $^*p<.05$.

(∼20%), but we note it is often strongly correlated with other, likely important factors we did not control for, such as data quality, which disproportionately hurts low-resource languages (Oladipo et al., 2023; Wang et al., 2025)[10] (see § Limitations). Script is a further possible confound, since SOV and SVO languages differ systematically in writing system: in FLORES-200, SOV languages are non-alphabetic far more often than SVO (71% vs. 19%). Restricting to segmental alphabets, however, the gap still grows with data (29 SVO / 7 SOV: near zero at 5 MB to $+0.057$ at 1 GB, 95% CI $[+0.011, +0.131]$). Beyond the aggregate BPEC results, per-constituent surprisal further confirms that preferences are data-driven: the ranking reverses from S < O < V in semantic-free artificial to V < O < S in natural languages both on FLORES-200 and PUD data, with the same architecture yielding opposite profiles depending on whether semantic dependencies are present.

Multilingual Models The SVO advantage and the right-branching directionality preference both extend to XGLM and mGPT (Table 2; Appendix Figure 12), where SVO groups are dominated by highly-resourced Indo-European languages.[11] BLOOM is the exception: 21 of its 28 SVO languages are Niger-Congo languages, most very-low-resource, with a median training data volume of 1.7 MB, compared to 3.7 GB for all other languages[12], inflating the SVO group's median BPEC and masking the advantage. More broadly, SVO languages receive substantially more training data than SOV languages across all three model families: in BLOOM, the median SVO/SOV ratio is 11× (Laurençon et al., 2022), reduced to 5× in XGLM[13] after upsampling. The preference thus reflects resource scale rather than word order. All three model families (mGPT, XGLM, and BLOOM) were explicitly designed to broaden language coverage beyond English-centric corpora, yet the SVO advantage persists wherever the SVO group is not dominated by very-low-resource languages; in BLOOM, the wide interquartile range (IQR) of SVO languages (Table 2, Appendix Figure 12) reflects the heterogeneity between the high-resource non-Niger-Congo and the very-low-resource Niger-Congo languages within this group.

5.3 Comparison of Artificial and Natural Language Results

At small training scales, artificial and natural language results partially align: models show a left-branching advantage and no clear base word order preference in both settings. As data grows, however, the two settings consistently diverge across base word orders, internal constituent switches, and constituent-level surprisal trends.

The left-branching advantage is stable across data sizes in artificial languages but vanishes in natural languages at scale, where right-branching SVO languages become easiest. Base word order rankings also reverse; SVO is hardest in artificial languages at 10 MB (Appendix Figure 9) but easiest in natural languages at 1 GB (Table 1). Per-constituent surprisal also flips, consistent with semantic selectional restrictions reshaping predictability in natural languages (Futrell, 2019). These contrasts across artificial and natural languages establish that word order preferences in transformer LMs are shaped by data rather than architecture; our multilingual results further reinforce this, with the models' SVO preference disappearing only when very-low-resource languages dominate this word order. Even monolingual Goldfish models trained on content-equivalent amounts of data

[10] Within the SVO group of Goldfish models, the easiest language shifts with scale: Indonesian has the lowest BPEC at 5 and 10 MB, with English just behind, but English leads at larger scales, consistent with higher-resource languages benefiting more from cleaner data as data grows.

[11] On an alphabetic script subset the multilingual SVO advantage persists in mGPT (22 SVO / 10 SOV, SVO vs. SOV Mann–Whitney p=.027 at 1.3B, .030 at 13B; Cliff's $\delta\approx-0.50$). BLOOM and XGLM have too few alphabetic SOV languages (1–2) to test.

[12] The median for the remaining non-Niger-Congo SVO languages in BLOOM is 79 GB.

[13] https://huggingface.co/facebook/xglm-564M

per language (Arnett et al., 2024) favour the SVO order most commonly seen in highly-resourced languages at scale (Appendix A.4), suggesting that data quantity alone does not fully explain this preference. Instead, the SVO advantage arises from properties of the training data beyond word order: it survives controls for morphological complexity, script, and, in the coarse form we can measure, training data composition, with only language resourcedness partly explaining it (§5.2), which is itself a proxy for data quality (Oladipo et al., 2023; Wang et al., 2025); finer aspects of data quality we leave to future work (see § Limitations).

6 Conclusion

We present a systematic analysis of word order preferences in decoder LMs across artificial and natural languages at multiple training data scales, combining both to disentangle architectural biases from data-driven preferences. On artificial languages, models prefer left-branching configurations with no alignment with cross-linguistic universals, and SVO languages become hardest at scale; on natural languages, no preference emerges at small scales, but LMs exhibit a preference for SVO languages as data grows while SOV falls behind. These preferences are data-driven: the same architecture exhibits opposite rankings on artificial and natural languages, and the preference for SVO order in natural languages correlates with resource level and data quality rather than word order type. Crucially, even where architectural biases exist (e.g., the left-branching preference on artificial languages), they are overridden by data effects at scale.

Since highly-resourced languages are overwhelmingly SVO, this data-driven advantage has real typological consequences. Given that 1 GB of training data suffices for the SVO preference to emerge even in monolingual models, the effects we study are likely more pronounced in frontier models where the training data quantity and quality gaps between high- and low-resource languages are even larger. This creates a precondition for what we term *silent typological homogenisation*: a hypothesised gradual reduction of word order diversity in languages that productively use multiple base word orders towards that of the dominant languages. To confirm this hypothesis and better understand the implications of LM word order preferences, we argue that extending this analysis to larger models and spoken language data, testing whether finetuning or downstream task application alters word order preferences (Belinkov and Glass, 2019), and applying mechanistic interpretability to identify how the observed order preferences are encoded (Olah, 2022) remain important directions for future work.

Limitations

The scope of our experiments is necessarily limited. Our PCFG production rules, following White and Cotterell (2021) and Kuribayashi et al. (2024), cover a limited subset of the typological landscape: relative clause placement, for instance, cannot be fully captured by a binary switch given the range of attested strategies (Dryer and Haspelmath, 2013). Our experiments focus on the GPT-2 architecture, consistent with prior work (White and Cotterell, 2021; Kallini et al., 2024) and motivated by direct comparability with Goldfish models (Chang et al., 2026), which share the same architecture and hyperparameters. We did not evaluate larger multilingual models beyond mGPT, BLOOM, and XGLM due to insufficient transparency regarding per-language training data composition, which is essential for our analysis of resource-level effects. Our language sample may also be biased toward Indo-European languages, which are overrepresented in both typological databases and training data.

Our evaluation data and metrics also carry certain limitations. We evaluate on written, relatively formal corpora (FLORES-200, PUD), but spoken language shows substantially greater word order flexibility (Hüll and Dobrovoljc, 2025), and register and genre strongly influence constituent ordering, so our results may not generalise to spoken or informal registers. FLORES-200 and PUD consist of translated text, which may introduce translationese biases (Koppel and Ordan, 2011). Metrics such as PPL and BPEC, while standard, measure overall compression rather than word order preferences directly; behavioural analysis of model outputs would provide more direct evidence.

Our data quality analysis also relies on the observed correlation between language resourcedness (Joshi et al., 2020, which we find the SVO advantage tracks) and data quality. While higher-resource languages tend to have cleaner, more diverse corpora (Kreutzer et al., 2022; Oladipo et al., 2023; Wang et al., 2025), and web-crawled data is noisiest for low-resource languages (Kreutzer et al., 2022), our training data composition variable (OS-

CAR web-crawl share) remains a coarse proxy, as we do not have direct annotations for data genre or quality. Isolating which finer aspects of data quality drive the effect requires per-language annotation of register, source, and corpus cleanliness, which we leave to future work.

Finally, our analysis relies on categorical word order labels from databases such as WALS (Dryer and Haspelmath, 2013) and Grambank (Skirgård et al., 2023), which encode features in simplified, often binary ways. WALS feature 81A classifies dominant word order based on main clause order alone, with no representation of variability across clause types or the pragmatic conditions under which ordering alternations arise. This can introduce noise: Modern Standard Arabic is classified as VSO based on a 1958 source, whereas more recent work documents flexible VSO/SVO alternation (Arad Greshler et al., 2017; Himmelreich, 2023); German instead is classified as having no dominant order in WALS despite arguably SVO-dominant main clauses, the very criterion WALS uses for classification, while exhibiting SOV in subordinate clauses and clause-final non-finite verbs in auxiliary and modal constructions. Similarly, Russian is classified as SVO-dominant and Belarusian as having no dominant order in WALS, despite comparable word order flexibility in UD corpora (Nivre et al., 2020). Word order variation is arguably gradient rather than categorical (Levshina et al., 2023), and a single label may obscure the flexibility of a language's actual word order distribution; even languages labeled with a single dominant order contain sentences with alternative orderings in the evaluation data. Resources such as APiCS partially address this by encoding word order frequencies, but the underlying data are themselves difficult to obtain: reliable estimates require representative corpora, consistent annotation standards across languages, and large-scale annotation by linguists with near-native competence, none of which is straightforward to achieve. More broadly, many typological features are not binary: Afrikaans, for example, has prepositions, postpositions, and circumpositions.

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A Appendix

A.1 Artificial Language Details

Vocabulary Source words were drawn from the most frequent English words in the `wordfreq` library (Speer, 2022) and used to generate pseudowords with Wuggy (Keuleers and Brysbaert, 2010). The resulting vocabulary contains ∼50k wordforms across five open-class categories: nouns inflecting for number (50%), adjectives (10%), and three verb classes (intransitive, transitive, complement-taking) inflecting for number and tense (40% combined). Closed-class items comprise 4 prepositions, 7 pronouns, and single invariant tokens for the subordinating complementiser, relativiser, coordinating conjunction, and overt subject and object case markers. This POS inventory follows White and Cotterell (2021) and subsequent work (Kuribayashi et al., 2024; El-Naggar et al., 2025b); fully replicating natural language POS complexity is infeasible within a controlled PCFG setup. The noun-heavy distribution (50%) reflects this restricted inventory and aligns

with corpus-based POS statistics (Hudson, 1994). Within each open-class category, items are assigned Zipfian sampling weights (α=1.0) by frequency rank; closed-class items are sampled uniformly.

Training Corpus Each language variant was used to generate 5 MB and 10 MB of training text by sampling sentences from the PCFG, compared to 10,000 sentences in White and Cotterell (2021). The PCFG allows recursive expansion, with a maximum of 400 expansions per sentence to bound recursion depth, following White and Cotterell (2021). Generated sentences have an average length of 12 tokens.

Tokeniser We train a shared SentencePiece BPE tokeniser (Kudo and Richardson, 2018) (15k vocabulary) on the generated corpus in word-boundary mode, so merges never cross word boundaries. Because all 192 languages are permutations of the same sentences with identical vocabulary and word frequencies, the subword inventory depends only on the shared word list, not on word order: retraining the tokeniser separately on each grammar yields identical 15,000-piece vocabularies. We also enable `add_dummy_prefix` so that sentence-initial words receive the same token IDs as mid-sentence occurrences, avoiding position-based bias. To confirm our results are not tokeniser-driven, we additionally retrained all 192 models (single seed) with a unigram tokeniser (as used in the Goldfish models; Chang et al., 2026) and obtained comparable results: the difficulty ordering is nearly unchanged (Spearman ρ=0.94 at 5 MB, 0.91 at 10 MB), all five switch effects keep their direction, and the base-order extremes are preserved.

Model and Hyperparameters We fully reproduced the results of White and Cotterell (2021) before conducting our experiments. We then made the following modifications. We adopt the Goldfish architecture (Chang et al., 2026) to match the model size used in our natural language experiments, enabling direct comparison (full hyperparameters in Table 3). Our models match the Goldfish configuration in layers, attention heads, and embedding dimensions, differing only in vocabulary size (15k vs 50k); prior work with a much smaller vocabulary (∼1,400 pseudowords; White and Cotterell, 2021; Kuribayashi et al., 2024) recovers the same trends, so our results are not vocabulary-size-driven. We set weight decay to 0, as the default value of 0.01 led to over-regularisation at our model size and data volume, roughly doubling perplexity in pilot experiments. We reduce warmup steps to 10% of total training steps. We cap training at 20 epochs rather than the 10,000 update steps used by White and Cotterell (2021), which on their data volume results in over 100 epochs. In pilot experiments on 5 MB and 10 MB corpora, we observed train–validation loss crossover at epochs ∼10 and ∼16, respectively, with validation loss plateauing or increasing beyond epoch 50. Limiting training to 20 epochs is consistent with Muennighoff et al. (2023), who find diminishing gains beyond 16 epochs of data repetition; with Chang et al. (2026), who train Goldfish models for 10 epochs; and with Kuribayashi et al. (2024), who use 10 epochs in a similar setup with a smaller training corpus. We evaluate at step level and select the checkpoint with the lowest validation loss. We limit artificial language experiments to 5 and 10 MB to match the Goldfish architecture at these sizes; at larger scales, Goldfish uses more layers, precluding direct comparison. Artificial languages are also easier to model than natural.

Hyperparameter	**Value**
Model	
Architecture	GPT-2 (decoder-only)
Layers	4
Attention heads	8
Hidden size	512
FFN inner size	2,048
Dropout	0.1
Activation	ReLU
Tied embeddings	Yes
Parameters	20.6M
Tokeniser	
Type	SentencePiece BPE (word-boundary)
Vocabulary size	15,000
Data	
Train / Dev / Test	80 / 10 / 10
Optimisation	
Framework	HuggingFace Transformers
Optimiser	Adam (β_1=0.9, β_2=0.999, ε=1e-6)
Learning rate	1e-4
LR schedule	Linear decay with warmup
Warmup	10% of total steps
Weight decay	0
Gradient clipping	1.0
Sequence length	512 tokens
Batch size	64 sequences (4 × 16 grad. accum.)
Max epochs	20
Precision	bf16

Table 3: Model and training hyperparameters for artificial language experiments.

Compute Cost For artificial languages, we trained 1,920 models (192 grammars $\times$ 10 splits) at each of two data sizes (5 MB and 10 MB) on NVIDIA A100 GPUs. A single model takes approximately 2.5 minutes (5 MB) and 5 minutes (10 MB), totalling approximately 252 A100 GPU-hours with a wall-clock time of approximately 2 days using parallelised Slurm array jobs.

A.2 Natural Language Details

Word Order Labeling We map each switch in our artificial grammar (Figure 2) to attested typological features: **Base word order**: WALS 81A; **Switch 1** (complement clause order): Grambank GB135; **Switch 2** (complementiser position): Grambank GB421; **Switch 3** (adposition order): WALS 85A; **Switch 4** (adjective–noun order): WALS 87A, Grambank GB193; **Switch 5** (relative clause–noun order): WALS 90A. Features were drawn from `lang2vec` (Littell et al., 2017), which aggregates WALS, SSWL, and Ethnologue. Where `lang2vec` lacked coverage, we consulted WALS (Dryer and Haspelmath, 2013) directly, then Grambank (Skirgård et al., 2023) and APiCS (Michaelis et al., 2013) via URIEL+ (Khan et al., 2025). FLORES-200 covers 200 languages, with script alternatives for four (204 total). Of these, 52 lacked a base word order label in WALS, Grambank, and APiCS; we supplemented these using peer-reviewed articles for specific languages, corroborated where possible by corpus-based labels from UD treebanks (Nivre et al., 2020), bringing coverage to 194 languages. Of the 194, 132 have binary (L/R) values for all five switches. Languages with non-binary or missing values were excluded from the corresponding analysis.

Evaluation Languages Tables 4–5 list all 103 FLORES-200 languages with a base word order label that appear in at least one evaluation set and form the basis of all analyses in §5: 68 Goldfish languages available at all four training sizes,[14] 46 BLOOM, 30 XGLM, and 56 mGPT training languages; 20 of the 103 also have a PUD treebank.

Goldfish Model Coverage on FLORES-200 We evaluated Goldfish models on 183 of FLORES-200 languages, requiring a strict one-to-one match between FLORES codes and dedicated Goldfish models; we do not adopt the cross-variety substitutions in Chang et al. (2026) (e.g. `awa_Deva` $\rightarrow$ `hin_Deva`), as word order differences between varieties could confound our analysis. The 21 excluded varieties (`ace_Arab`, `acm_Arab`, `acq_Arab`, `aeb_Arab`, `ajp_Arab`, `arb_Latn`, `ars_Arab`, `ary_Arab`, `awa_Deva`, `bjn_Arab`, `kam_Latn`, `kas_Arab`, `min_Arab`, `mni_Beng`, `npi_Deva`, `nus_Latn`, `ory_Orya`, `sat_Beng`, `taq_Latn`, `taq_Tfng`, `tzm_Tfng`) lack a dedicated model due to missing script coverage or absence from Goldfish entirely.

A.3 Evaluation

BPEC For artificial languages, raw perplexity is directly comparable because all 192 variants share the same vocabulary, tokeniser, and derivation probabilities. For natural languages, per-token PPL is not comparable across scripts and tokenisers. We therefore follow Cotterell et al. (2018) and normalise corpus negative log-likelihood by the English character count of the parallel corpus to obtain bits per English character (BPEC):

$$\mathrm{BPEC}_\ell = \frac{-\sum_{s \in S_\ell} \sum_{t=1}^{n_s} \log_2 p_\theta\big(x_t^{(s)} \mid x_{<t}^{(s)}\big)}{C_{\mathrm{eng}}} \quad (1)$$

where S_ℓ are the evaluation sentences in language ℓ, n_s the scored subword tokens per sentence, and C_{eng} the total character count of the English portion of the respective corpus (FLORES-200/PUD). We prepend the model's start token (`[CLS]` for Goldfish, `[BOS]` for multilingual models) as conditioning context, but exclude it from the scored tokens.

Greenbergian Universals From Greenberg's 45 universals (Greenberg, 1963), we select the subset testable with syntactic constructions in our artificial languages. Since our PCFG generates only declarative sentences, we are limited to universals involving base S/V/O order, adpositions, adjective–noun order, and relative clauses. This yields four implicational universals: Universal 1 (subject precedes object in the dominant orders SOV, SVO, VSO), Universal 3 (VSO languages are prepositional), Universal 4 (SOV languages are postpositional), and Universal 17 (VSO languages place the adjective after the noun). We also test Dryer's observation that OV languages correlate with prenominal relative clauses and VO languages with postnominal ones (Dryer, 1992). For each universal, we

[14] Larger sets are available at individual sizes (e.g., 130 at 5 MB) but are not used in §5 to ensure that the same languages are compared across all four training scales, enabling attribution to data volume rather than language sample changes.

compare mean test PPL between the configuration predicted by the universal and its counterpart.

A.4 Language Resource Level and Word Order Distribution

We use the resource taxonomy of Joshi et al. (2020), which classifies languages into six classes (0–5) based on the availability of labeled and unlabeled NLP data, as a proxy for resourcedness (column *Res.* in Tables 4–5). We report statistics on the 68 Goldfish languages listed in Tables 4–5, tracked across all four training scales, as these are the languages for which we directly observe the emerging SVO advantage (65 with a resource class assignment in Joshi et al. (2020)). SVO languages have substantially higher resourcedness than SOV (mean resource class 3.35 vs. 2.24; Mann–Whitney U=472, p=0.003, r=0.72). Of the 19 highly-resourced (Class 4–5) {SVO, SOV} languages, 14 are SVO (74%), and a Cochran–Armitage trend test confirms that the proportion of SVO rises monotonically with resource class (z=3.03, p=0.002).

A.5 Mixed-Effects Analysis of the SVO–SOV Gap

In line with prior work that uses mixed-effects models to separate language-modeling difficulty from confounding language properties (Mielke et al., 2019), we fit linear mixed-effects models to the Goldfish BPEC of the 50 SVO/SOV languages (32 SVO, 18 SOV) in our full directionality-annotated set (Figure 1c,d), across all four training sizes (200 language-by-size observations; Table 6); VSO and NoDominant languages also shown in Figure 1c,d are excluded, as the model contrasts SVO against SOV. The model is

$$\texttt{bpec} \sim \texttt{is_SOV} \times \texttt{size} + \text{covariates} + (1 \mid \texttt{language}),$$

with training size entered as z-scored $\log_{10}$(MB), so main effects are read at the centre of the size range (∼47 MB) and interactions per SD of log-size (∼8× data). The covariates (all reported in Table 6) are morphological complexity (`mattr_z`), language resourcedness (`resource_lvl`, z-scored resourcedness class from Joshi et al., 2020), and training data composition (`data_comp`, OSCAR web-crawl share).[15] We measure morphological complexity as subword MATTR (1000-token window) under each language's own tokeniser (as in Tatariya et al., 2025), since corpus-based measures such as TTR have been shown to correlate with, and be interchangeable with, typological complexity measures derived from databases such as WALS (Bentz et al., 2016), and it is available for all languages, so it does not limit the sample. Models are fit by maximum likelihood so the nested models M1–M9 are comparable; the robustness block in Table 6 refits key models with a maximal random-effects structure (random intercept + random size slope per language). Our finding is the `is_SOV` × `size` interaction (the growth of the SVO–SOV performance gap with training data scale), and each covariate is tested as a predictor of it, both as a level term and as an interaction. Table 6 reports all coefficients across the nine nested models (M1–M9) and the robustness checks. Our finding, the `is_SOV`×`size` interaction, survives every control: it stays significant across M2–M9 and under the maximal random-effects structure, shrinking only ∼20% (from +0.049 to +0.038) when resourcedness is allowed to scale with data (M7, M9). The main effect (`is_SOV`, the average SVO–SOV gap), by contrast, is absorbed: it is significant on its own (M1) and with any single covariate (M2–M4) but non-significant once all three are added together (M5), so the covariates jointly explain the average gap, but not its growth with data. Among the covariates, only resourcedness has a significant standalone size interaction (`resource_lvl`×`size`, β=−0.012, p=.037; M7), in the same direction as our effect: higher-resource languages improve faster with data, widening the gap for the lower-resourced SOV group, which accounts for the ∼20% reduction. Morphological complexity does not act differently on SOV (`mattr_z`×`is_SOV`, p=.39; M6) and its own effect shrinks with data (within SVO, MATTR × log-size β=−0.017, p=.014), so it cannot explain an effect that grows with data. Training data composition shows no size interaction on its own (`data_comp`×`size`, p=.098; M8); in the full model (M9) it is collinear with resourcedness (r=0.64), so neither size interaction is individually significant there, though the word order interaction is unchanged.

A.6 Additional Results

Additional artificial language results appear in Figures 8–10; natural language results in Figures 6, 7,

[15] The share of each language's Goldfish training data drawn from OSCAR rather than more curated corpora, computed from the per-language `proportions` column of the Goldfish data documentation: https://github.com/tylerachang/goldfish/blob/main/data/goldfish_data_info.tsv.

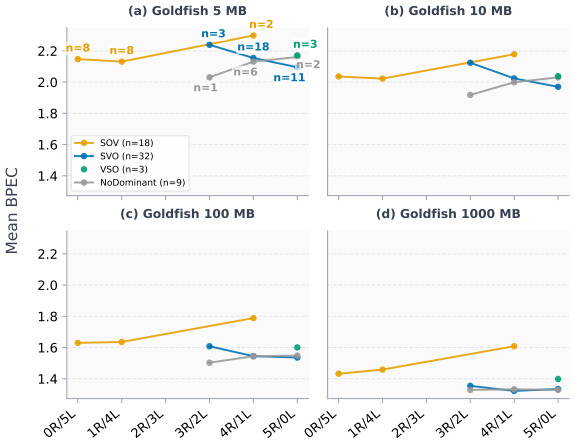


Figure 6: Directionality vs. BPEC at all four Goldfish scales, same 62 languages (cf. Figure 1c–d for 5 and 1 GB only). SVO separates from SOV at 100 MB–1 GB.

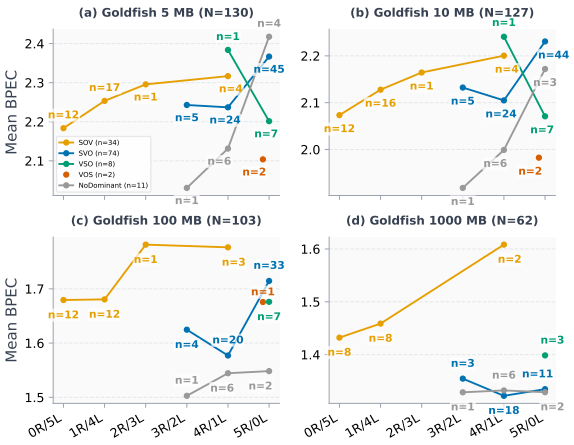


Figure 7: As Figure 6 but with all available languages per scale. Same pattern, ruling out selection artefacts.

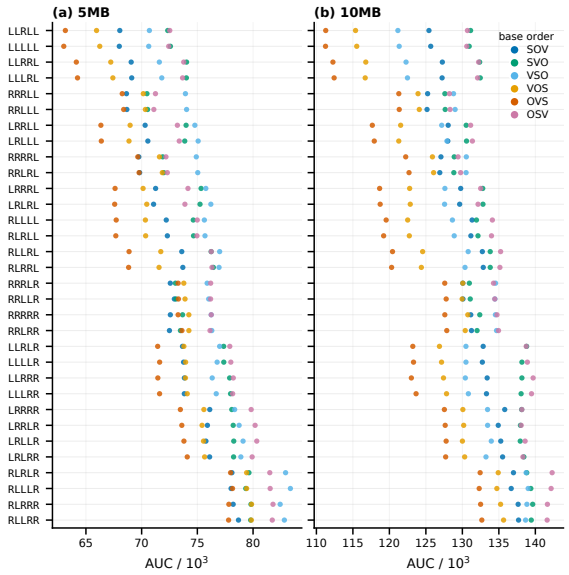


Figure 8: Convergence AUC ($\div 10^3$; lower = faster) for all 32 artificial language configurations at 5 MB (a) and 10 MB (b), ranked by cross-order mean. Rankings stable across sizes (Spearman $\rho = 0.884$) and strongly correlated with final PPL ($\rho > 0.89$).

11, 12, and Table 7.

A.7 Artefact Use and Licensing

All artefacts are used for research consistent with their intended purpose. FLORES-200 (NLLB Team et al., 2022) and PUD (Zeman et al., 2017): CC-BY-SA 4.0; WALS (Dryer and Haspelmath, 2013) and Grambank (Skirgård et al., 2023): CC-BY 4.0; BLOOM (BigScience Workshop et al., 2023): BigScience RAIL v1.0; XGLM (Lin et al., 2022): MIT; mGPT (Shliazhko et al., 2024) and Stanza (Qi et al., 2020): Apache 2.0; Wuggy (Keuleers and Brysbaert, 2010): GPL; wordfreq (Speer, 2022): MIT. Goldfish models (Chang et al., 2026) are publicly available but unlicensed.

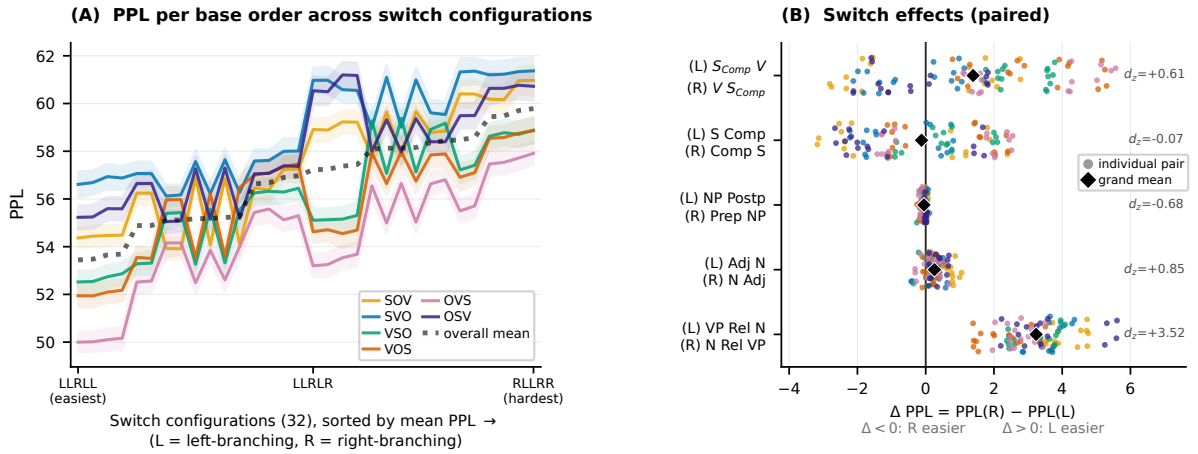


Figure 9: Replication of Figure 4 with 10 MB training samples. Setup and notation are identical. All five paired t-tests remain significant after Holm–Bonferroni correction ($p < .001$).

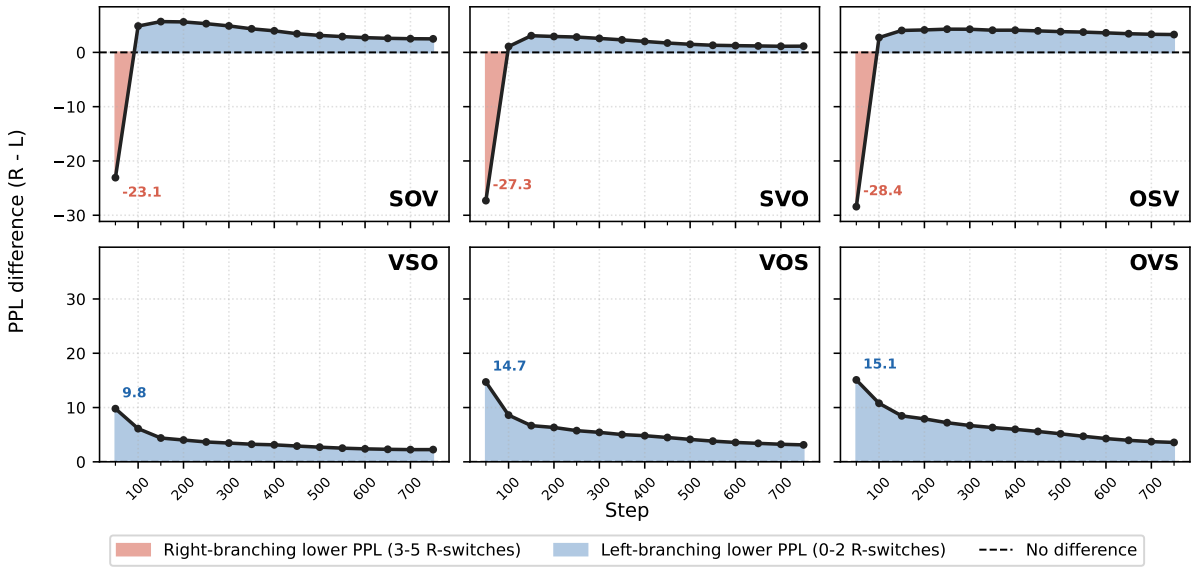


Figure 10: Extended version of Figure 5 showing all six base orders. SV orders: right-branching initially easier, flips after ∼100 steps. VS orders: left-branching preferred throughout. Same trend at 10 MB.

Language	FLORES	Label	Family	Res.	Model				
					Goldfish	BLOOM	XGLM	mGPT	PUD
Afrikaans	afr_Latn	NoDom-RRRLR	Indo-European	3	✓	×	×	✓	×
Akan	aka_Latn	SVO-RNLR?	Niger-Congo	1	×	✓*	×	×	×
Amharic	amh_Ethi	SOV-LLRLL	Afro-Asiatic	2	✓	×	×	×	×
Armenian	hye_Armn	NoDom-RRLLN	Indo-European	1	✓	×	×	✓	×
Assamese	asm_Beng	SOV-L??L?	Indo-European	1	×	✓*	×	×	×
Ayacucho Quechua	quy_Latn	SOV-L?LLL	Quechuan	–	×	×	✓*	×	×
Bambara	bam_Latn	SOV-LLLRL	Niger-Congo	1	×	✓	×	×	×
Bashkir	bak_Cyrl	SOV-LRLLL	Turkic	1	×	×	×	✓	×
Basque	eus_Latn	SOV-LLLRL	Isolate	4	✓	✓	✓	✓	×
Belarusian	bel_Cyrl	SVO-RRRLR	Indo-European	3	✓	×	×	✓	×
Bengali	ben_Beng	SOV-LRLLL	Indo-European	3	✓	✓	✓	✓	×
Bosnian	bos_Latn	SVO-RRRLR	Indo-European	3	✓	×	×	×	×
Bulgarian	bul_Cyrl	SVO-RRRLR	Indo-European	3	✓	×	✓	✓	×
Burmese	mya_Mymr	SOV-LNLRL	Sino-Tibetan	1	×	×	✓	✓	×
Catalan	cat_Latn	SVO-RRRRR	Indo-European	4	✓	✓	✓	×	×
Chinese (Simp.)	zho_Hans	SVO-RRRLL	Sino-Tibetan	5	✓	✓	✓	×	✓
Chinese (Trad.)	zho_Hant	SVO-RRRLL	Sino-Tibetan	5	×	✓	×	×	×
Croatian	hrv_Latn	SVO-RRRLR	Indo-European	4	✓	×	×	×	×
Czech	ces_Latn	SVO-RRRLR	Indo-European	4	✓	×	×	×	✓
Danish	dan_Latn	SVO-RRRLR	Indo-European	3	✓	×	×	✓	×
Dutch	nld_Latn	NoDom-RRRLR	Indo-European	4	✓	×	×	×	×
Eastern Panjabi	pan_Guru	SOV-RRLLN	Indo-European	2	✓	✓	×	×	×
English	eng_Latn	SVO-RRRLR	Indo-European	5	✓	✓	✓	✓	✓
Esperanto	epo_Latn	NoDom-RRRLR	Constructed	1	✓	×	×	×	×
Estonian	est_Latn	SVO-RRLLR	Uralic	3	✓	×	✓	✓	×
Finnish	fin_Latn	SVO-RRLLR	Uralic	4	✓	×	✓	✓	✓
Fon	fon_Latn	SVO-RRNRR	Niger-Congo	0	×	✓	×	×	×
French	fra_Latn	SVO-RRRRR	Indo-European	5	✓	✓	✓	✓	✓
Galician	glg_Latn	NoDom-RRRRR	Indo-European	3	✓	×	×	×	×
Ganda	lug_Latn	SVO-RRRRR	Niger-Congo	1	×	✓	×	×	×
Georgian	kat_Geor	NoDom-RRLLR	Kartvelian	3	✓	×	×	✓	×
German	deu_Latn	NoDom-RRRLR	Indo-European	5	✓	×	✓	✓	✓
Greek	ell_Grek	NoDom-RRRLR	Indo-European	3	✓	×	✓	✓	×
Gujarati	guj_Gujr	SOV-LRLLL	Indo-European	1	✓	✓	×	×	×
Haitian Creole	hat_Latn	SVO-RRRRR	Creole	0	×	×	✓	×	×
Halh Mongolian	khk_Cyrl	SOV-LLLLL	Mongolic	–	×	×	×	✓	×
Hausa	hau_Latn	SVO-RRRLR	Afro-Asiatic	2	✓	×	×	×	×
Hebrew	heb_Hebr	SVO-RRRRR	Afro-Asiatic	3	✓	×	×	✓	×
Hindi	hin_Deva	SOV-LRLLL	Indo-European	4	✓	✓	✓	✓	✓
Hungarian	hun_Latn	NoDom-RRLLN	Uralic	4	✓	×	×	✓	×
Icelandic	isl_Latn	SVO-RRRLR	Indo-European	2	✓	×	×	×	✓
Igbo	ibo_Latn	SVO-RRRRR	Niger-Congo	1	×	✓	×	×	×
Indonesian	ind_Latn	SVO-RRRRR	Austronesian	3	✓	✓	✓	✓	✓
Italian	ita_Latn	NoDom-RRRRR	Indo-European	4	✓	×	✓	✓	✓
Japanese	jpn_Jpan	SOV-LLLLL	Japonic	5	✓	×	✓	✓	✓
Javanese	jav_Latn	SVO-RRRRR	Austronesian	1	×	×	×	✓	×
Kannada	kan_Knda	SOV-LLLLL	Dravidian	1	✓	✓	×	×	×
Kazakh	kaz_Cyrl	SOV-L????	Turkic	3	✓*	×	×	✓*	×
Kikuyu	kik_Latn	SVO-RRRRR	Niger-Congo	1	×	✓	×	×	×
Kinyarwanda	kin_Latn	SVO-R?RR?	Niger-Congo	1	×	✓*	×	×	×
Korean	kor_Hang	SOV-LLLLL	Koreanic	4	✓	×	✓	✓	✓
Kyrgyz	kir_Cyrl	SOV-L?LL?	Turkic	1	✓*	×	×	✓*	×

Table 4: Evaluation languages (part 1 of 2): the 103 FLORES-200 languages with a base word order label appearing in at least one evaluation set. *Label*: dominant order + five head-direction parameters following the encoding in Figure 2 (NoDom = No Dominant Order). *Res.*: resource class 0–5 (Joshi et al., 2020); – = not classified. *Model* columns: presence in training data (Goldfish = available at all four training sizes, 68/103; BLOOM, 46; XGLM, 30; mGPT, 56). *PUD*: PUD treebank available (20/103; all PUD languages fall within this set). ✓*: included for base word order analysis but excluded from directionality analysis (§5) due to missing parameters (?). N = no dominant order.

Language	FLORES	Label	Family	Res.	Model				PUD
					Goldfish	BLOOM	XGLM	mGPT	
Lingala	lin_Latn	SVO-R?RRR	Niger-Congo	1	×	✓*	×	×	×
Lithuanian	lit_Latn	NoDom-RRRLR	Indo-European	3	✓	×	×	✓	×
Macedonian	mkd_Cyrl	SVO-RRRLR	Indo-European	1	✓	×	×	×	×
Malayalam	mal_Mlym	SOV-LLLLL	Dravidian	1	✓	✓	×	✓	×
Maltese	mlt_Latn	SVO-RRRRR	Afro-Asiatic	2	✓	×	×	×	×
Marathi	mar_Deva	SOV-RNLLL	Indo-European	2	✓	✓	×	✓	×
MSA	arb_Arab	VSO-RRRRR	Afro-Asiatic	–	✓	✓	✓	✓	✓
Nepali	npi_Deva	SOV-LLLL?	Indo-European	1	×	✓*	×	×	×
Northern Sotho	nso_Latn	SVO-R????	Niger-Congo	1	×	✓*	×	×	×
Northern Uzbek	uzn_Latn	SOV-LRLLL	Turkic	–	×	×	×	✓	×
Norwegian Bokmål	nob_Latn	SVO-RRRLR	Indo-European	1	✓	×	×	×	×
Nyanja	nya_Latn	SVO-RRRRR	Niger-Congo	1	×	✓	×	×	×
Odia	ory_Orya	SOV-?NLLN	Indo-European	1	×	✓*	×	×	×
Polish	pol_Latn	SVO-RRRLR	Indo-European	4	✓	×	×	✓	✓
Portuguese	por_Latn	SVO-RRRRR	Indo-European	4	✓	✓	✓	✓	✓
Romanian	ron_Latn	SVO-RRRRR	Indo-European	3	✓	×	×	✓	×
Rundi	run_Latn	SVO-RRRRR	Niger-Congo	0	×	✓	×	×	×
Russian	rus_Cyrl	SVO-RRRLR	Indo-European	4	✓	×	✓	✓	✓
Serbian	srp_Cyrl	SVO-RRRLR	Indo-European	4	✓	×	×	×	×
Shona	sna_Latn	SVO-RRRRR	Niger-Congo	1	×	✓	×	×	×
Sinhala	sin_Sinh	SOV-LLLLL	Indo-European	0	✓	×	×	×	×
Slovak	slk_Latn	SVO-RRRLR	Indo-European	3	✓	×	×	×	×
Slovenian	slv_Latn	SVO-RRRLR	Indo-European	3	✓	×	×	×	×
Somali	som_Latn	SOV-LRNRR	Afro-Asiatic	1	✓	×	×	×	×
Southern Sotho	sot_Latn	SVO-RRRRR	Niger-Congo	1	×	✓	×	×	×
Spanish	spa_Latn	SVO-RRRRR	Indo-European	5	✓	✓	✓	✓	✓
Std. Latvian	lvs_Latn	SVO-RRRLR	Indo-European	–	×	×	×	✓	×
Std. Malay	zsm_Latn	SVO-RRRRR	Austronesian	–	×	×	×	✓	×
Swahili	swh_Latn	SVO-RRRRR	Niger-Congo	–	✓	✓	✓	✓	×
Swedish	swe_Latn	SVO-RRRLR	Indo-European	4	✓	×	×	✓	✓
Tagalog	tgl_Latn	VSO-RRRNR	Austronesian	3	✓	×	×	×	×
Tajik	tgk_Cyrl	SOV-LRRRR	Indo-European	1	✓	×	×	×	×
Tamil	tam_Taml	SOV-LLLLL	Dravidian	3	✓	✓	✓	×	×
Tatar	tat_Cyrl	SOV-LRLLL	Turkic	1	✓	×	×	✓	×
Telugu	tel_Telu	SOV-LLLLL	Dravidian	1	✓	✓	✓	✓	×
Thai	tha_Thai	SVO-RRRRR	Tai-Kadai	3	✓	×	✓	✓	✓
Tsonga	tso_Latn	SVO-RR?R?	Niger-Congo	1	×	✓*	×	×	×
Tswana	tsn_Latn	SVO-RRRRR	Niger-Congo	2	×	✓	×	×	×
Tumbuka	tum_Latn	SVO-R????	Niger-Congo	1	×	✓*	×	×	×
Turkish	tur_Latn	SOV-LLLLL	Turkic	4	✓	×	✓	✓	✓
Turkmen	tuk_Latn	SOV-LLLLL	Turkic	1	×	×	×	✓	×
Twi	twi_Latn	SVO-RNLR?	Niger-Congo	1	×	✓*	×	×	×
Ukrainian	ukr_Cyrl	SVO-RRRLR	Indo-European	3	✓	×	×	✓	×
Urdu	urd_Arab	SOV-LRLLL	Indo-European	3	✓	✓	✓	×	×
Vietnamese	vie_Latn	SVO-RRRRR	Austroasiatic	4	✓	✓	✓	✓	×
Welsh	cym_Latn	VSO-RRRRR	Indo-European	1	✓	×	×	×	×
Western Persian	pes_Arab	SOV-?RRRR	Indo-European	–	✓*	×	×	✓*	×
Wolof	wol_Latn	SVO-RRRRR	Niger-Congo	2	×	✓	×	×	×
Xhosa	xho_Latn	SVO-RRRRR	Niger-Congo	2	×	✓	×	×	×
Yoruba	yor_Latn	SVO-RRRRR	Niger-Congo	2	×	✓	×	✓	×
Zulu	zul_Latn	SVO-RRRRR	Niger-Congo	2	×	✓	×	×	×

Table 5: Evaluation languages (part 2 of 2, continued from Table 4). Column definitions are identical. MSA = Modern Standard Arabic.

Term	β	SE	p
M1 base (no covariates)			
is_SOV	**+0.071**	0.024	**.003**
is_SOV×size	**+0.049**	0.011	**<.001**
M2 + morphological complexity (level)			
is_SOV	**+0.050**	0.024	**.041**
is_SOV×size	**+0.049**	0.011	**<.001**
mattr_z	**+0.027**	0.012	**.021**
M3 + language resourcedness (level)			
is_SOV	**+0.053**	0.025	**.033**
is_SOV×size	**+0.049**	0.011	**<.001**
resource_lvl	−0.023	0.012	.057
M4 + training data composition (level)			
is_SOV	**+0.052**	0.025	**.034**
is_SOV×size	**+0.049**	0.011	**<.001**
data_comp	−0.021	0.012	.076
M5 + all three levels			
is_SOV	+0.030	0.025	.224
is_SOV×size	**+0.049**	0.011	**<.001**
mattr_z	**+0.025**	0.011	**.022**
resource_lvl	−0.011	0.014	.441
data_comp	−0.014	0.014	.317
M6 + morphological complexity × word order			
is_SOV	**+0.055**	0.025	**.029**
is_SOV×size	**+0.049**	0.011	**<.001**
mattr_z	**+0.034**	0.014	**.018**
mattr_z×is_SOV	−0.022	0.025	.386
M7 + language resourcedness × size			
is_SOV	+0.036	0.025	.147
is_SOV×size	**+0.039**	0.012	**<.001**
resource_lvl×size	−0.012	0.006	**.037**
mattr_z	**+0.024**	0.012	**.037**
M8 + training data composition × size			
is_SOV	+0.032	0.025	.202
is_SOV×size	**+0.040**	0.012	**<.001**
data_comp	−0.021	0.011	.064
data_comp×size	−0.009	0.006	.098
mattr_z	**+0.027**	0.011	**.015**
M9 full model (resourcedness & composition × size)			
is_SOV	+0.030	0.025	.224
is_SOV×size	**+0.038**	0.012	**.001**
mattr_z	**+0.025**	0.011	**.022**
resource_lvl×size	−0.010	0.007	.165
data_comp×size	−0.003	0.007	.634
Robustness: maximal random effects			
M1: is_SOV×size	**+0.049**	0.011	**<.001**
M7: is_SOV×size	**+0.039**	0.012	**.001**
M9: is_SOV×size	**+0.038**	0.012	**.002**
binary res. is_SOV×size	**+0.037**	0.011	**.001**
binary res. high_res×size	−0.032	0.012	**.007**

Table 6: Mixed-effects models of Goldfish BPEC (50 SVO/SOV languages × 4 sizes). **Bold** = $p < .05$. Each covariate (morphological complexity, resourcedness, and training-data composition, OSCAR share) is entered as a level term (M2–M4), jointly (M5), and in interaction (M6–M9). Our finding, the word order × size interaction (is_SOV×size, the SOV gap growing with data), survives every control (M2–M9) and a maximal random-effects structure. The main effect (is_SOV, the SVO–SOV gap at the average training size, ∼47 MB) is absorbed once all three covariate levels are added together (M5).

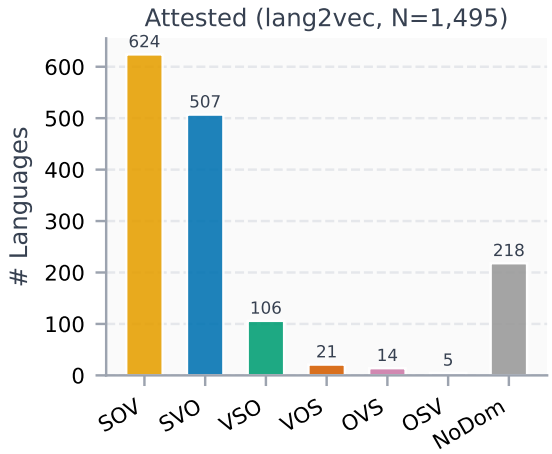


Figure 11: Base word order distribution across attested languages. Figure 1a uses only the N=526 subset with complete switch annotations for all five parameters, which skews toward well-documented, right-branching Indo-European languages.

Order	n	5 MB	10 MB	100 MB	1 GB
SVO	13	2.20	2.08	**1.60**	**1.38**
SOV	4	**2.16**	**2.05**	1.67	1.49
VSO	1	2.26	2.15	1.72	1.50
NoDominant	2	2.21	2.08	1.61	1.38

Table 7: Median BPEC of Goldfish models on PUD treebank languages by base word order. SOV leads at small scales; SVO leads at large scales, mirroring FLORES-200 (Table 1). Per-order samples are small (n=1–13), so we omit IQR and test the ordering directly: the SVO advantage reaches significance by 1 GB (SVO vs. SOV Mann–Whitney p=.045), and the SOV×size interaction replicates FLORES-200 (mixed-effects β=+0.04, p=.04).

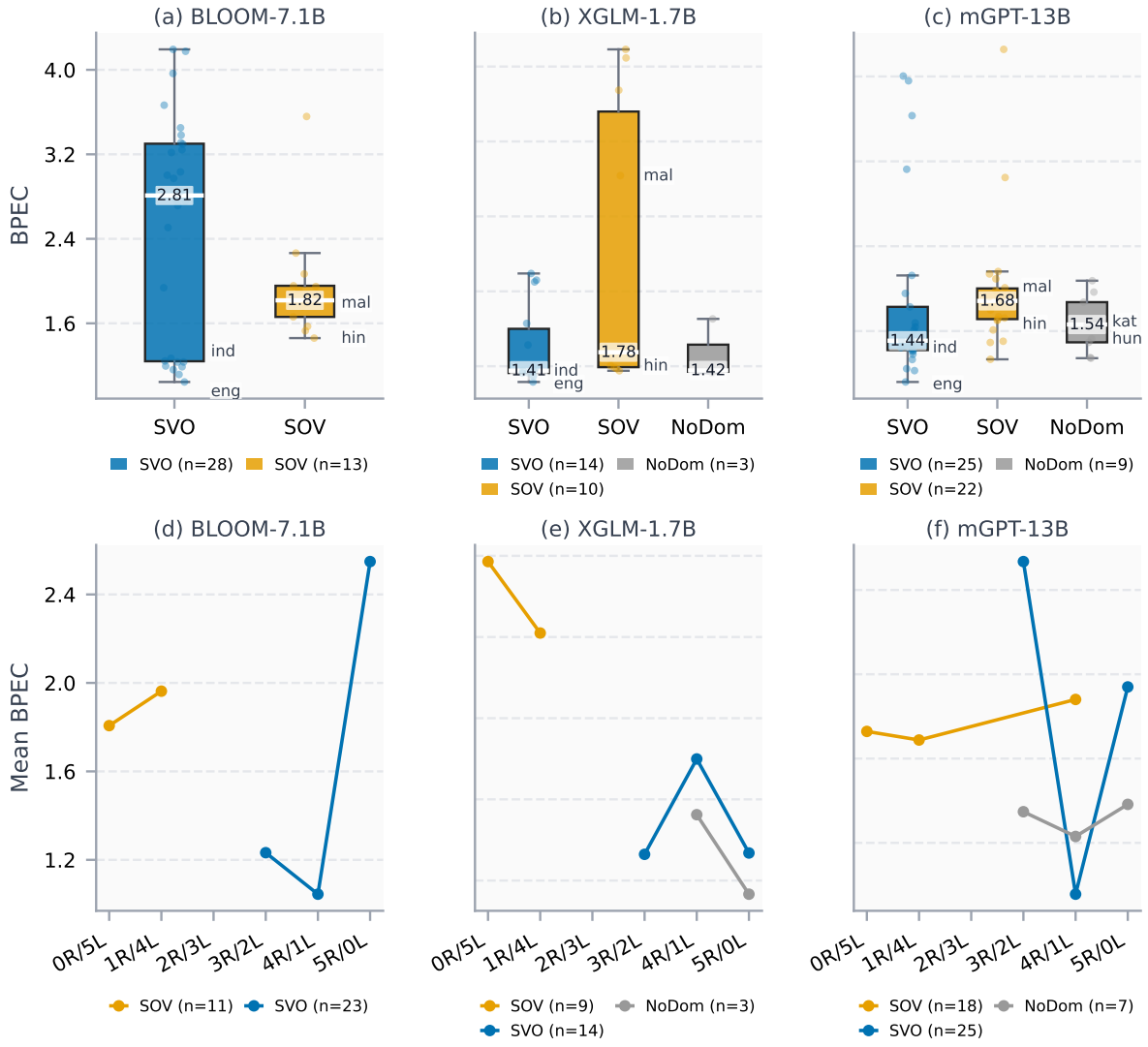


Figure 12: BPEC by base word order (a–c) and head directionality (d–f) for the largest model in each multilingual family (FLORES-200); smaller sizes show the same patterns. SVO compresses better than SOV in XGLM and mGPT; BLOOM reverses because 21 of 28 SVO languages are very-low-resource Niger-Congo languages (§5.2). The right-branching SVO advantage is visible even in BLOOM's directionality panels (d–f), where it is masked at the base word order level.